

**Q1: What is a "System" in modeling and simulation?**

- A) Computer program
- B) Real world object to study ✓
- C) Math equation
- D) Drawing

**Q2: What is a "Model"?**

- A) Real system
- B) Simplified representation ✓
- C) Computer only
- D) Random data

**Q3: What is "Simulation"?**

- A) Building models
- B) Imitating real system over time ✓
- C) Collecting data
- D) Writing equations

**Q4: Bank example - What is the "System"?**

- A) Flow diagram
- B) Computer program
- C) Real bank with customers ✓
- D) Queue formula

**Q5: Bank example - What is the "Model"?**

- A) Bank building
- B) Tellers
- C) Customer flow diagram ✓
- D) Waiting line

**Q6: Bank example - What is the "Simulation"?**

- A) Real customers
- B) Bank manager
- C) Program showing queue growth ✓
- D) Deposit slips

**Q7: Model used when math approach is possible?**

- A) Simulation
- B) Experimental
- C) Analytic ✓
- D) Physical

**Q8: Model used for complex systems?**

- A) Analytic
- B) Simulation ✓
- C) Experimental
- D) Simple

**Q9: Model used when real system exists?**

- A) Simulation
- B) Analytic
- C) Experimental ✓
- D) Theoretical

**Q10: Result that tells what system will do?**

- A) Explanation
- B) Design
- C) Prediction ✓
- D) Validation

**Q11: Result that tells why system behaves as it does?**

- A) Prediction
- B) Explanation ✓
- C) Design
- D) Measurement

**Q12: Result intended to achieve a purpose?**

- A) Prediction
- B) Explanation
- C) Design ✓
- D) Analysis

**Q13: Comparing real measurements with simulation results?**

- A) Prediction
- B) Design
- C) Validation ✓
- D) Modeling

**Q14: What does "what if" questions help with?**

- A) Building banks
- B) Investigating system changes ✓
- C) Writing code
- D) Drawing diagrams

**Q15: When is it better to do simulation?**

- A) After implementation
- B) During implementation
- C) Before implementation ✓
- D) Never

**Q18: Which software is professional simulation tool mentioned?**

- A) Word
- B) Excel
- C) AnyLogic ✓
- D) PowerPoint

**Q19: When is simulation appropriate for risky situations?**

- A) Always
- B) Never
- C) When experimentation may damage system ✓
- D) Only for small systems

**Q20: When is simulation NOT appropriate?**

- A) Complex systems
- B) Problem solved by common sense ✓
- C) Expensive experiments
- D) Risky situations

**Q21: When is simulation NOT appropriate?**

- A) System too complex
- B) Problem solved analytically ✓
- C) Environmental changes
- D) New designs

**Q22: Advantage of simulation - time can be?**

- A) Only increased
- B) Only decreased
- C) Compressed or expanded ✓
- D) Ignored

**Q23: Advantage of simulation - helps understand?**

- A) Only outputs
- B) Variable interactions ✓
- C) Only inputs
- D) Nothing

**Q24: Disadvantage of simulation?**

- A) Cheap
- B) Fast
- C) Requires special training ✓
- D) Easy to interpret

**Q25: Disadvantage of simulation?**

- A) Results easy to understand
- B) Results difficult to interpret ✓
- C) Always accurate
- D) Quick to build

**Q28: What is an "Entity" in a system?**

- A) Property of object
- B) Object of interest ✓
- C) Time period
- D) System state

**Q29: What is an "Attribute"?**

- A) Object of interest
- B) Property of entity ✓
- C) Time period
- D) System change

**Q30: What is an "Activity"?**

- A) Object of interest
- B) Entity property
- C) Time period of specified length ✓
- D) System state